

Alan Amadeus Davila

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Education

Texas A&M University — College Station, TX

B.S. in Electrical & Computer Engineering | Expected Graduation: December 2026

Certification: LinkedIn Learning — *Become a PLC Developer* Certificate, 2026

Projects

Automated Conveyor Control System — PLC & HMI

Independent Project, 2026 – Present

Ladder Logic, OpenPLC, Docker, FUXA HMI

- Developed PLC ladder logic for conveyor **start/stop control**, **motor seal-in logic**, and **Manual/Automatic operating modes** using OpenPLC.
- Implementing sensor-based jam detection, alarm handling, and HMI status visualization to identify conveyor faults and support operator troubleshooting.
- Integrating a FUXA-based virtual HMI with conveyor controls, status indicators, sensor feedback, and fault/alarm functionality for operator interaction.

Arduino UNO R4 WiFi Closed-Loop Drawer Controller [\[GitHub\]](#)

Independent Project, 2026

C++, Arduino, Quadrature Encoders, IBT-2 Motor Drivers, Limit Switches

- Designed an upgraded **Arduino UNO R4 WiFi control architecture** using dual quadrature encoders and four end-of-travel limit switches for position feedback and boundary detection.
- Developed **C++ closed-loop positioning logic** to command each drawer to 0%, 50%, or 100% based on encoder counts instead of fixed motor run times.
- Implemented automatic homing, endpoint protection, and safety timeouts to establish reference positions and prevent mechanical overtravel.

Raspberry Pi 4 Automated Drawer System [\[GitHub\]](#)

Independent Project, 2025

GT2 Belt drives, DC Motors, IBT-2 Motor Drivers, Limit Switches

- Designed and assembled a dual-drawer electromechanical system using **GT2 Belt Drives**, DC motors, IBT-2 motor drivers, push-button inputs, and limit switches.
- Developed Python GPIO control logic for simultaneous drawer operation with timed **closed, half-open, and fully open positions**.
- Implemented **simultaneous dual-motor control** with multi-click push-button commands, allowing both drawers to operate independently without blocking each other.

Skills

PLC & Controls: Ladder Logic, OpenPLC, FUXA HMI, PLC sequencing, interlocks, motor control, PID fundamentals

Languages: C, C++, Python, JavaScript, SQL, SystemVerilog, Verilog

Hardware & Test: DC motors, H-bridge drivers, analog/digital circuits, PCB troubleshooting, oscilloscopes, DMMs

Facilities Systems: BAS exposure, AggieWorks DB, work-order verification, building-system troubleshooting coordination

Recent Work Experience

Texas A&M University - College Station

August 2022 – Present

Junior Facilities Coordinator

College Station, TX

- Inspected completed campus maintenance work orders using AggieWorks DB, verifying contractor/internal work completion and documenting deficiencies requiring follow-up service.
- Coordinated with university facilities staff, SCC personnel, third-party contractors, and BAS/facilities personnel to track repair status and support timely completion of building maintenance tasks.